

MATH 301: Homework 5

Aren Vista

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Section 3.4 : 6 Let $x_n := n^{1/n}$ for $n \in \mathbb{N}$.

a) Show that $x_{n+1} < x_n$ IFF $(1 + 1/n)^n < n$, and infer that the inequality is valid for $3 \leq n$
(See Example 3.3.6) Conclude that x_n is ultimately decreasing and that $\exists \lim x_n = L$

b) Use the fact that the subsequence x_{2n} also converges to L to conclude that $x = 1$

Subproblem a)

Case $(x_{n+1} < x_n \rightarrow (1 + 1/n)^n < n)$ Suppose $x_{n+1} < x_n$

Case $((1 + 1/n)^n < n \rightarrow x_{n+1} < x_n)$

Subproblem b)